

CSA4302-INTERNET PROGRAMMING (IP)

ASSESSMENT 3

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Scenario: Web Development Standards Implementation

Question 1: Select the correct HTML tag for video embedding and justify your answer.

Answer:

Correct Option: B. `<video src="video.mp4" controls></video>`

Justification:

The `<video>` tag is the standard HTML5 element used to embed videos on a webpage. The controls attribute provides built-in controls such as **Play**, **Pause**, **Volume**, and **Fullscreen**.

Example:

```
<video src="video.mp4" controls></video>
```

Why the other options are incorrect:

- **A. `<media>`** – Not a valid HTML5 tag.
- **C. `<movie>`** – Not a valid HTML element.
- **D. `<vid>`** – Not supported in HTML.

Question 2: Identify the appropriate HTTP method for form submission and explain why.

Answer:

Correct Option: B. POST

Justification:

The **POST** method securely sends user data in the request body instead of the URL. It is commonly used for registration forms, login forms, and other operations involving sensitive information.

Example:

```
<form action="/register" method="POST">
```

Why the other options are incorrect:

- **A. GET** – Sends data through the URL and is not secure for sensitive information.

- **C. PUT** – Used mainly to update existing resources.
- **D. DELETE** – Used to remove resources from the server.

Question 3: Choose the proper semantic element for navigation and justify its usage.

Answer:

Correct Option: B. <nav>

Justification:

The <nav> element defines a navigation section that contains links to different pages or sections of a website. It improves accessibility, readability, and search engine optimization (SEO).

Example:

```
<nav>

  <a href="index.html">Home</a>

  <a href="about.html">About</a>

  <a href="contact.html">Contact</a>

</nav>
```

Why the other options are incorrect:

- **A. <div>** – Generic container without semantic meaning.
- **C. <section>** – Used to group related content, not specifically for navigation.
- **D. <header>** – Represents introductory content, not navigation alone.

Question 4: Identify the correct XML syntax and explain the error in other options.

Answer:

Correct Option:

B.<note>

```
<to>User</to>

<from>Admin</from>

</note>
```

Justification:

This XML is well-formed because:

- All tags are properly opened and closed.
- Tags are correctly nested.
- There is a single root element (<note>).

Errors in the other options:

- **Option A**

<note><to>User</to><from>Admin</from>

Error: Missing closing </note> tag.

- **Option C**

<note><to>User</from></note>

Error: Mismatched closing tag (</from> instead of </to>).

- **Option D**

<note to="User"></note>

Question 5: Choose the correct protocol for web communication and justify your choice.

Answer:

Correct Option: B. HTTP

Justification:

HTTP (HyperText Transfer Protocol) is the standard protocol used for communication between web browsers and web servers. It enables users to request and receive web pages over the Internet.

Why the other options are incorrect:

- **A. FTP** – Used for file transfer.
- **C. SMTP** – Used for sending emails.
- **D. SNMP** – Used for network monitoring and management.